Operator product expansion, Riemannian manifolds and Laplacians

Yi-Zhi Huang

Department of Mathematics
Rutgers University
Piscataway, NJ 08854, USA
and
Beijing International Center for Mathematical Research
Peking University, Beijing, China

June 11, 2012

Chern Institute of Mathematics, Nankai University

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models

- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models

- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Operator product expansion

- In quantum theory, physical quantities are not numbers anymore. They are operators acting on Hilbert spaces.
- A quantum field theory is described by quantum fields, which are operator-valued functions on space-time.
- Operator product expansion: Let A and B be quantum fields and $A(x)$ and $B(y)$ their values at x and y. Then there is an open neighborhood O of y such that for all x in $O \setminus \{y\}$,

$$A(x)B(y) = \sum_i d_i(x-y)C^i(y)$$

where C_i are quantum fields and d_i are analytic functions on $O \setminus \{y\}$.

Operator product expansion

- In quantum theory, physical quantities are not numbers anymore. They are operators acting on Hilbert spaces.
- A quantum field theory is described by quantum fields, which are operator-valued functions on space-time.
- Operator product expansion: Let A and B be quantum fields and $A(x)$ and $B(y)$ their values at x and y. Then there is an open neighborhood O of y such that for all x in $O \setminus \{y\}$,

$$A(x)B(y) = \sum_i d_i(x-y)C^i(y)$$

where C_i are quantum fields and d_i are analytic functions on $O \setminus \{y\}$.

Operator product expansion

- In quantum theory, physical quantities are not numbers anymore. They are operators acting on Hilbert spaces.
- A quantum field theory is described by quantum fields, which are operator-valued functions on space-time.
- Operator product expansion: Let A and B be quantum fields and $A(x)$ and $B(y)$ their values at x and y. Then there is an open neighborhood O of y such that for all x in $O \setminus \{y\}$,

$$A(x)B(y) = \sum_i d_i(x-y)C^i(y)$$

where C_i are quantum fields and d_i are analytic functions on $O \setminus \{y\}$.

Examples: Vertex operator algebras

Belavin-Polyakov-Zamolodchikov, Borcherds, Frenkel-Lepowsky-Meurman.

A vertex operator algebra consists the following data:

- A $\mathbb{Z}$-graded vector space $V = \coprod_{n\in\mathbb{Z}} V_{(n)}$.
- A vertex operator map

$$\begin{aligned} Y_V : V \otimes V &\to V[[z, z^{-1}]], \\ u \otimes v &\mapsto Y(u, z)v. \end{aligned}$$

- A vacuum $\mathbf{1} \in V_{(0)}$.

Examples: Vertex operator algebras

Belavin-Polyakov-Zamolodchikov, Borcherds, Frenkel-Lepowsky-Meurman.

A vertex operator algebra consists the following data:

- A $\mathbb{Z}$-graded vector space $V = \coprod_{n\in\mathbb{Z}} V_{(n)}$.
- A vertex operator map

$$\begin{aligned} Y_V : V \otimes V &\to V[[z, z^{-1}]], \\ u \otimes v &\mapsto Y(u, z)v. \end{aligned}$$

- A vacuum $\mathbf{1} \in V_{(0)}$.

Examples: Vertex operator algebras

Belavin-Polyakov-Zamolodchikov, Borcherds, Frenkel-Lepowsky-Meurman.

A vertex operator algebra consists the following data:

- A $\mathbb{Z}$-graded vector space $V = \coprod_{n\in\mathbb{Z}} V_{(n)}$.
- A vertex operator map

$$\begin{aligned} Y_V : V \otimes V &\to V[[z, z^{-1}]], \\ u \otimes v &\mapsto Y(u, z)v. \end{aligned}$$

- A vacuum $\mathbf{1} \in V_{(0)}$.

Examples: Vertex operator algebras

Belavin-Polyakov-Zamolodchikov, Borcherds, Frenkel-Lepowsky-Meurman.

A vertex operator algebra consists the following data:

- A $\mathbb{Z}$-graded vector space $V = \coprod_{n\in\mathbb{Z}} V_{(n)}$.
- A vertex operator map

$$\begin{aligned} Y_V : V \otimes V &\to V[[z, z^{-1}]], \\ u \otimes v &\mapsto Y(u, z)v. \end{aligned}$$

- A vacuum $\mathbf{1} \in V_{(0)}$.

Examples: Vertex operator algebras

Belavin-Polyakov-Zamolodchikov, Borcherds, Frenkel-Lepowsky-Meurman.

A vertex operator algebra consists the following data:

- A $\mathbb{Z}$-graded vector space $V = \coprod_{n\in\mathbb{Z}} V_{(n)}$.
- A vertex operator map

$$\begin{aligned} Y_V : V \otimes V &\to V[[z, z^{-1}]], \\ u \otimes v &\mapsto Y(u, z)v. \end{aligned}$$

- A vacuum $\mathbf{1} \in V_{(0)}$.

Examples: Vertex operator algebras

These data satisfy the following axioms:

- Grading-restriction property: $\dim V_{(n)} < \infty$ for $n \in \mathbb{Z}$ and $V_{(n)} = 0$ when n is sufficiently negative.
- Lower-truncation property: For $u, v \in V$, $Y(u, z)v$ contains only finitely many negative power terms.
- Axioms for the vacuum: For $u \in V$, $Y(\mathbf{1}, z)u = u$ and $\lim_{z \to 0} Y(u, z)\mathbf{1} = u$.
- Axioms for the conformal element: Let $L(n) : V \to V$ be defined by $Y(\omega, z) = \sum_{n \in \mathbb{Z}} L(n) z^{-n-2}$, then $[L(m), L(n)] = (m - n)L(m + n) + \frac{c}{12}(m^3 - m)\delta_{m+b,0}$, $\frac{d}{dz} Y(u, z) = Y(L(-1)u, z)$ ($L(-1)$-derivative property) for $u \in V$ and $L(0)u = nu$ for $u \in V_{(n)}$ ($L(0)$-grading property).

Examples: Vertex operator algebras

These data satisfy the following axioms:

- Grading-restriction property: $\dim V_{(n)} < \infty$ for $n \in \mathbb{Z}$ and $V_{(n)} = 0$ when n is sufficiently negative.
- Lower-truncation property: For $u, v \in V$, $Y(u, z)v$ contains only finitely many negative power terms.
- Axioms for the vacuum: For $u \in V$, $Y(\mathbf{1}, z)u = u$ and $\lim_{z \to 0} Y(u, z)\mathbf{1} = u$.
- Axioms for the conformal element: Let $L(n) : V \to V$ be defined by $Y(\omega, z) = \sum_{n \in \mathbb{Z}} L(n) z^{-n-2}$, then $[L(m), L(n)] = (m - n)L(m + n) + \frac{c}{12}(m^3 - m)\delta_{m+b,0}$, $\frac{d}{dz} Y(u, z) = Y(L(-1)u, z)$ ($L(-1)$-derivative property) for $u \in V$ and $L(0)u = nu$ for $u \in V_{(n)}$ ($L(0)$-grading property).

Examples: Vertex operator algebras

These data satisfy the following axioms:

- Grading-restriction property: $\dim V_{(n)} < \infty$ for $n \in \mathbb{Z}$ and $V_{(n)} = 0$ when n is sufficiently negative.
- Lower-truncation property: For $u, v \in V$, $Y(u, z)v$ contains only finitely many negative power terms.
- Axioms for the vacuum: For $u \in V$, $Y(\mathbf{1}, z)u = u$ and $\lim_{z \to 0} Y(u, z)\mathbf{1} = u$.
- Axioms for the conformal element: Let $L(n) : V \to V$ be defined by $Y(\omega, z) = \sum_{n \in \mathbb{Z}} L(n) z^{-n-2}$, then $[L(m), L(n)] = (m - n)L(m + n) + \frac{c}{12}(m^3 - m)\delta_{m+b,0}$, $\frac{d}{dz} Y(u, z) = Y(L(-1)u, z)$ ($L(-1)$-derivative property) for $u \in V$ and $L(0)u = nu$ for $u \in V_{(n)}$ ($L(0)$-grading property).

Examples: Vertex operator algebras

These data satisfy the following axioms:

- Grading-restriction property: $\dim V_{(n)} < \infty$ for $n \in \mathbb{Z}$ and $V_{(n)} = 0$ when n is sufficiently negative.
- Lower-truncation property: For $u, v \in V$, $Y(u, z)v$ contains only finitely many negative power terms.
- Axioms for the vacuum: For $u \in V$, $Y(\mathbf{1}, z)u = u$ and $\lim_{z \to 0} Y(u, z)\mathbf{1} = u$.
- Axioms for the conformal element: Let $L(n) : V \to V$ be defined by $Y(\omega, z) = \sum_{n \in \mathbb{Z}} L(n) z^{-n-2}$, then $[L(m), L(n)] = (m - n)L(m + n) + \frac{c}{12}(m^3 - m)\delta_{m+b,0}$, $\frac{d}{dz} Y(u, z) = Y(L(-1)u, z)$ ($L(-1)$-derivative property) for $u \in V$ and $L(0)u = nu$ for $u \in V_{(n)}$ ($L(0)$-grading property).

Examples: Vertex operator algebras

These data satisfy the following axioms:

- Grading-restriction property: $\dim V_{(n)} < \infty$ for $n \in \mathbb{Z}$ and $V_{(n)} = 0$ when n is sufficiently negative.
- Lower-truncation property: For $u, v \in V$, $Y(u, z)v$ contains only finitely many negative power terms.
- Axioms for the vacuum: For $u \in V$, $Y(\mathbf{1}, z)u = u$ and $\lim_{z \to 0} Y(u, z)\mathbf{1} = u$.
- Axioms for the conformal element: Let $L(n) : V \to V$ be defined by $Y(\omega, z) = \sum_{n \in \mathbb{Z}} L(n) z^{-n-2}$, then $[L(m), L(n)] = (m - n)L(m + n) + \frac{c}{12}(m^3 - m)\delta_{m+b,0}$, $\frac{d}{dz} Y(u, z) = Y(L(-1)u, z)$ ($L(-1)$-derivative property) for $u \in V$ and $L(0)u = nu$ for $u \in V_{(n)}$ ($L(0)$-grading property).

Examples: Vertex operator algebras

- Duality property: For $u_1, u_2, v \in V$, $v' \in V' = \coprod_{n\in\mathbb{Z}} V^*_{(n)}$, the series

$$\langle v', Y(u_1, z_1)Y(u_2, z_2)v\rangle$$
$$\langle v', Y(u_2, z_2)Y(u_1, z_1)v\rangle$$
$$\langle v', Y(Y(u_1, z_1 - z_2)u_2, z_2)v\rangle$$

  are absolutely convergent in the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$ and $|z_2| > |z_1 - z_2| > 0$, respectively, to a common rational function in z_1 and z_2 with the only possible poles at $z_1, z_2 = 0$ and $z_1 = z_2$.

Examples: Vertex operator algebras

In the region $|z_1| > |z_2| > |z_1 - z_2| > 0$, we have

$$Y(u_1, z_1)Y(u_2, z_2) = Y(Y(u_1, z_1 - z_2)u_2, z_2)$$

in a suitable sense. Write $Y(u_1, x)u_2 = \sum_{n\in\mathbb{Z}} Y_n(u_1)u_2 x^{-n-1}$. Then we have

$$Y(u_1, z_1)Y(u_2, z_2) = \sum_{n\in\mathbb{Z}} (z_1 - z_2)^{-n-1} Y(Y_n(u_1)u_2, z_2),$$

where $Y(u_1, z_1)$ and $Y(u_2, z_2)$ are the values of the quantum fields $Y(u_1, z)$ and $Y(u_2, z)$ at z_1 and z_2, respectively, $Y(Y_n(u_1)u_2, z_2)$ for $n \in \mathbb{Z}$ are the values of the quantum fields $Y(Y_n(u_1)u_2, z)$ at z_2 and $(z_1 - z_2)^{-n-1}$ for $n \in \mathbb{Z}$ are analytic functions of $z_1 - z_2$.

Examples: Modules for a vertex operator algebra

- Let V be a vertex operator algebra. A V-module is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{(n)}$ equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all those axioms for V which still make sense.
- A lower-bounded generalized V-module is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{(n)}$ equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all the axioms for a V-module except that $W_{(n)}$ are generalized eigenspaces instead of eigenspaces of $L(0)$ and are not necessarily finite dimensional. A weak V-module is a vector space W equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all those axioms for V-modules which still make sense.
- Vertex operators for these variants of modules have operator product expansion.

Examples: Modules for a vertex operator algebra

- Let V be a vertex operator algebra. A V-module is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{(n)}$ equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all those axioms for V which still make sense.
- A lower-bounded generalized V-module is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{(n)}$ equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all the axioms for a V-module except that $W_{(n)}$ are generalized eigenspaces instead of eigenspaces of $L(0)$ and are not necessarily finite dimensional. A weak V-module is a vector space W equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all those axioms for V-modules which still make sense.
- Vertex operators for these variants of modules have operator product expansion.

Examples: Modules for a vertex operator algebra

- Let V be a vertex operator algebra. A V-module is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{(n)}$ equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all those axioms for V which still make sense.
- A lower-bounded generalized V-module is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{(n)}$ equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all the axioms for a V-module except that $W_{(n)}$ are generalized eigenspaces instead of eigenspaces of $L(0)$ and are not necessarily finite dimensional. A weak V-module is a vector space W equipped with a vertex operator map $Y_W : V \otimes W \to W[[z, z^{-1}]]$ satisfying all those axioms for V-modules which still make sense.
- Vertex operators for these variants of modules have operator product expansion.

Examples: Intertwining operators

- Let W_1, W_2 and W_3 be V-modules. An intertwining operator of type $\binom{W_3}{W_1 W_2}$ is a linear map $\mathcal{Y} : W_1 \otimes W_2 \to W_3\{z\}$, where $W_3\{z\}$ is the space of all series in complex powers of z with coefficients in W_3, satisfying all those axioms for V which still make sense, that is, a lower-truncation property, an $L(-1)$-derivative property and a duality property.
- Question: Do we have operator product expansion for intertwining operators?
- Answer: Yes, under some conditions.

Examples: Intertwining operators

- Let W_1, W_2 and W_3 be V-modules. An intertwining operator of type $\binom{W_3}{W_1 W_2}$ is a linear map $\mathcal{Y} : W_1 \otimes W_2 \to W_3\{z\}$, where $W_3\{z\}$ is the space of all series in complex powers of z with coefficients in W_3, satisfying all those axioms for V which still make sense, that is, a lower-truncation property, an $L(-1)$-derivative property and a duality property.
- Question: Do we have operator product expansion for intertwining operators?
- Answer: Yes, under some conditions.

Examples: Intertwining operators

- Let W_1, W_2 and W_3 be V-modules. An intertwining operator of type $\binom{W_3}{W_1 W_2}$ is a linear map $\mathcal{Y} : W_1 \otimes W_2 \to W_3\{z\}$, where $W_3\{z\}$ is the space of all series in complex powers of z with coefficients in W_3, satisfying all those axioms for V which still make sense, that is, a lower-truncation property, an $L(-1)$-derivative property and a duality property.
- Question: Do we have operator product expansion for intertwining operators?
- Answer: Yes, under some conditions.

Examples: Intertwining operators

Theorem (H. 2002)

Let V be a simple vertex operator algebra satisfying the following conditions:

1. $V_{(n)} = 0$ *for* $n < 0$, $V_{(0)} = \mathbb{C}\mathbf{1}$ *and* V' *is isomorphic to* V *as a* V*-module.*
2. *Every lower-bounded generalized* V*-module is a direct sum of irreducible* V*-modules.*
3. V *is* C_2*-cofinite, that is,* $\dim V/C_2(V) < \infty$ *where* $C_2(V)$ *is the subspace of* V *spanned by elements of the form* $\mathrm{Res}_z z^{-2} Y(u,z)v$ *for* $u, v \in V$*. (Actually this condition can be replaced by a weaker condition.)*

Examples: Intertwining operators

Theorem (H. 2002)

Let V be a simple vertex operator algebra satisfying the following conditions:

1. $V_{(n)} = 0$ *for* $n < 0$, $V_{(0)} = \mathbb{C}\mathbf{1}$ *and* V' *is isomorphic to* V *as a* V*-module.*
2. *Every lower-bounded generalized* V*-module is a direct sum of irreducible* V*-modules.*
3. V *is* C_2*-cofinite, that is,* $\dim V/C_2(V) < \infty$ *where* $C_2(V)$ *is the subspace of* V *spanned by elements of the form* $\mathrm{Res}_z z^{-2} Y(u,z)v$ *for* $u, v \in V$*. (Actually this condition can be replaced by a weaker condition.)*

Examples: Intertwining operators

Theorem (H. 2002)

Let V be a simple vertex operator algebra satisfying the following conditions:

1. $V_{(n)} = 0$ *for* $n < 0$, $V_{(0)} = \mathbb{C}\mathbf{1}$ *and* V' *is isomorphic to* V *as a* V*-module.*
2. *Every lower-bounded generalized* V*-module is a direct sum of irreducible* V*-modules.*
3. V *is* C_2*-cofinite, that is,* $\dim V/C_2(V) < \infty$ *where* $C_2(V)$ *is the subspace of* V *spanned by elements of the form* $\mathrm{Res}_z z^{-2} Y(u,z)v$ *for* $u, v \in V$*. (Actually this condition can be replaced by a weaker condition.)*

Examples: Intertwining operators

Theorem (H. 2002)

Let V be a simple vertex operator algebra satisfying the following conditions:

1. $V_{(n)} = 0$ *for* $n < 0$, $V_{(0)} = \mathbb{C}\mathbf{1}$ *and* V' *is isomorphic to* V *as a* V*-module.*
2. *Every lower-bounded generalized* V*-module is a direct sum of irreducible* V*-modules.*
3. V *is* C_2*-cofinite, that is,* $\dim V/C_2(V) < \infty$ *where* $C_2(V)$ *is the subspace of* V *spanned by elements of the form* $\mathrm{Res}_z z^{-2} Y(u,z)v$ *for* $u, v \in V$*. (Actually this condition can be replaced by a weaker condition.)*

Examples: Intertwining operators

Then the intertwining operators satisfy an associativity property, that is, given intertwining operators $\mathcal{Y}_1$ and $\mathcal{Y}_2$, there exists intertwining operators $\mathcal{Y}_3$ and $\mathcal{Y}_4$ such that for elements w_1 and w_2 of suitable modules and z_1 and z_2 satisfying $|z_1| > |z_2| > |z_1 - z_2| > 0$,

$$\mathcal{Y}_1(w_1, z_1)\mathcal{Y}_2(w_2, z_2) = \mathcal{Y}_3(\mathcal{Y}_4(w_1, z_1 - z_2)w_2, z_2).$$

Write

$$\mathcal{Y}_4(w_1, z) = \sum_{n\in\mathbb{R}}(\mathcal{Y}_4)_n(w_1)z^{-n-1}.$$

Then we have

$$\mathcal{Y}_1(w_1, z_1)\mathcal{Y}_2(w_2, z_2) = \sum_{n\in\mathbb{R}}(z_1 - z_2)^{-n-1}\mathcal{Y}_3((\mathcal{Y}_4)_n(w_1)w_2, z_2).$$

Examples: Intertwining operators

Then the intertwining operators satisfy an associativity property, that is, given intertwining operators $\mathcal{Y}_1$ and $\mathcal{Y}_2$, there exists intertwining operators $\mathcal{Y}_3$ and $\mathcal{Y}_4$ such that for elements w_1 and w_2 of suitable modules and z_1 and z_2 satisfying $|z_1| > |z_2| > |z_1 - z_2| > 0$,

$$\mathcal{Y}_1(w_1, z_1)\mathcal{Y}_2(w_2, z_2) = \mathcal{Y}_3(\mathcal{Y}_4(w_1, z_1 - z_2)w_2, z_2).$$

Write

$$\mathcal{Y}_4(w_1, z) = \sum_{n\in\mathbb{R}}(\mathcal{Y}_4)_n(w_1)z^{-n-1}.$$

Then we have

$$\mathcal{Y}_1(w_1, z_1)\mathcal{Y}_2(w_2, z_2) = \sum_{n\in\mathbb{R}}(z_1 - z_2)^{-n-1}\mathcal{Y}_3((\mathcal{Y}_4)_n(w_1)w_2, z_2).$$

Examples: Meromorphic open-string vertex algebras

A meromorphic open-string vertex algebra is a $\mathbb{Z}$-graded vector space $V = \coprod_{n\in\mathbb{Z}} V_{(0)}$, a vertex operator map

$$\begin{aligned} Y_V : V \otimes V &\to V[[z, z^{-1}]], \\ u \otimes v &\mapsto Y(u, z)v, \end{aligned}$$

and a vacuum $\mathbf{1} \in V_{(0)}$, satisfying the grading-restriction property, the lower-truncation property, the axioms for the vacuum and rationality and associativity. Here rationality and associativity mean that for $u_1, u_2, v \in V$, $v' \in V' = \coprod_{n\in\mathbb{Z}} V^*_{(n)}$, the series

$$\langle v', Y(u_1, z_1)Y(u_2, z_2)v\rangle$$
$$\langle v', Y(Y(u_1, z_1 - z_2)u_2, z_2)v\rangle$$

are absolutely convergent in the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively, to a common rational function in z_1 and z_2 with the only possible poles at $z_1, z_2 = 0$ and $z_1 = z_2$.

Examples: Meromorphic open-string vertex algebras

In general, the vertex operators for a meromorphic open-string vertex algebra do not satisfy the full duality property. Many important properties, for example, Jacobi identity, commutativity, locality, the commutator formula, skew-symmetry or even the associator formula, of vertex operator algebras are not satisfied by the vertex operators for a meromorphic open-string vertex algebra. On the other hand, the vertex operators for a meromorphic open-string vertex algebra still have operator product expansion.

Theorem (H. 2012)

Given a finite-dimensional vector space $\mathfrak{h}$, let $\hat{\mathfrak{h}}_- = \mathfrak{h} \otimes t^{-1}\mathbb{C}[t^{-1}]$. Then the tensor algebra $T(\hat{\mathfrak{h}}_-)$ of $\hat{\mathfrak{h}}_-$ has a natural structure of meromorphic open-string vertex algebra.

Examples: Meromorphic open-string vertex algebras

In general, the vertex operators for a meromorphic open-string vertex algebra do not satisfy the full duality property. Many important properties, for example, Jacobi identity, commutativity, locality, the commutator formula, skew-symmetry or even the associator formula, of vertex operator algebras are not satisfied by the vertex operators for a meromorphic open-string vertex algebra. On the other hand, the vertex operators for a meromorphic open-string vertex algebra still have operator product expansion.

Theorem (H. 2012)

Given a finite-dimensional vector space $\mathfrak{h}$*, let* $\hat{\mathfrak{h}}_- = \mathfrak{h} \otimes t^{-1}\mathbb{C}[t^{-1}]$*. Then the tensor algebra* $T(\hat{\mathfrak{h}}_-)$ *of* $\hat{\mathfrak{h}}_-$ *has a natural structure of meromorphic open-string vertex algebra.*

Examples: Left modules for meromorphic open-string vertex algebras

We also have the notion of left modules for a meromorphic open-string vertex algebra. The vertex operators for a left module have operator product expansion.

Theorem (H. 2012)

Given a finite-dimensional vector space $\mathfrak{h}$ and a module Λ for the tensor algebra $T(\mathfrak{h})$, the space $T(\hat{\mathfrak{h}}_-) \otimes \Lambda$ has a natural structure of left module for the meromorphic open-string vertex algebra $T(\hat{\mathfrak{h}}_-)$.

Examples: Left modules for meromorphic open-string vertex algebras

We also have the notion of left modules for a meromorphic open-string vertex algebra. The vertex operators for a left module have operator product expansion.

Theorem (H. 2012)

Given a finite-dimensional vector space $\mathfrak{h}$ and a module Λ for the tensor algebra $T(\mathfrak{h})$, the space $T(\hat{\mathfrak{h}}_-) \otimes \Lambda$ has a natural structure of left module for the meromorphic open-string vertex algebra $T(\hat{\mathfrak{h}}_-)$.

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Harmonic maps from a Riemann surface to a Riemnannian manifold

- Σ: 2-diemnsional Riemannian manifold.
  M: n-dimensional Riemannian manifold.
  φ: smooth map from Σ to M.
- Action: $\int_\Sigma \| d\varphi \| dS$.
  Locally: $\| d\varphi \| = \eta^{ij} g_{\mu\nu} \frac{\partial \varphi^\mu}{\partial x^i} \frac{\partial \varphi^\nu}{\partial x^j}$.
- Harmonic maps: Critical points of the action above.
- The action is conformally invariant. Only the Riemann surface structure on Σ is needed.

Harmonic maps from a Riemann surface to a Riemnannian manifold

- Σ: 2-diemnsional Riemannian manifold.
  M: n-dimensional Riemannian manifold.
  φ: smooth map from Σ to M.
- Action: $\int_\Sigma \parallel d\varphi \parallel dS$.
  Locally: $\parallel d\varphi \parallel = \eta^{ij} g_{\mu\nu} \frac{\partial \varphi^\mu}{\partial x^i} \frac{\partial \varphi^\nu}{\partial x^j}$.
- Harmonic maps: Critical points of the action above.
- The action is conformally invariant. Only the Riemann surface structure on Σ is needed.

Harmonic maps from a Riemann surface to a Riemnannian manifold

- Σ: 2-diemnsional Riemannian manifold.
  M: n-dimensional Riemannian manifold.
  φ: smooth map from Σ to M.
- Action: $\int_\Sigma \parallel d\varphi \parallel dS$.
  Locally: $\parallel d\varphi \parallel = \eta^{ij} g_{\mu\nu} \frac{\partial \varphi^\mu}{\partial x^i} \frac{\partial \varphi^\nu}{\partial x^j}$.
- Harmonic maps: Critical points of the action above.
- The action is conformally invariant. Only the Riemann surface structure on Σ is needed.

Harmonic maps from a Riemann surface to a Riemnannian manifold

- Σ: 2-diemnsional Riemannian manifold.
  M: n-dimensional Riemannian manifold.
  φ: smooth map from Σ to M.
- Action: $\int_\Sigma \parallel d\varphi \parallel dS$.
  Locally: $\parallel d\varphi \parallel = \eta^{ij} g_{\mu\nu} \frac{\partial \varphi^\mu}{\partial x^i} \frac{\partial \varphi^\nu}{\partial x^j}$.
- Harmonic maps: Critical points of the action above.
- The action is conformally invariant. Only the Riemann surface structure on Σ is needed.

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Free bosons: Quantization of linear sigma models

- Linear sigma models: M flat.
- $M = \mathbb{R}^n$ and $p \in \mathbb{R}^n$.
  $T_pM = T_p\mathbb{R}^n$: the tangent space of $M = \mathbb{R}^n$ at p.
- Heisenberg algebra: $\widehat{T_pM} = \widehat{T_pM}_- \oplus \widehat{T_pM}_0 \oplus \widehat{T_pM}_+$, where

$$\begin{aligned}
\widehat{T_pM}_- &= T_pM \otimes t^{-1}\mathbb{C}[t^{-1}], \\
\widehat{T_pM}_0 &= T_pM \otimes \mathbb{C}t^0 \oplus \mathbb{C}\mathbf{l} \simeq T_pM \oplus \mathbb{C}\mathbf{l}, \\
\widehat{T_pM}_+ &= TM \otimes t\mathbb{C}[t].
\end{aligned}$$

- Eigenfunctions $|\mathbf{k}\rangle = e^{ik_\mu x^\mu}$ of the Laplacian with eigenvalues $-k_\mu k^\mu$.

Free bosons: Quantization of linear sigma models

- Linear sigma models: M flat.
- $M = \mathbb{R}^n$ and $p \in \mathbb{R}^n$.
  $T_pM = T_p\mathbb{R}^n$: the tangent space of $M = \mathbb{R}^n$ at p.
- Heisenberg algebra: $\widehat{T_pM} = \widehat{T_pM}_- \oplus \widehat{T_pM}_0 \oplus \widehat{T_pM}_+$, where

$$\begin{aligned}
\widehat{T_pM}_- &= T_pM \otimes t^{-1}\mathbb{C}[t^{-1}], \\
\widehat{T_pM}_0 &= T_pM \otimes \mathbb{C}t^0 \oplus \mathbb{C}\mathbf{l} \simeq T_pM \oplus \mathbb{C}\mathbf{l}, \\
\widehat{T_pM}_+ &= TM \otimes t\mathbb{C}[t].
\end{aligned}$$

- Eigenfunctions $|\mathbf{k}\rangle = e^{ik_\mu x^\mu}$ of the Laplacian with eigenvalues $-k_\mu k^\mu$.

Free bosons: Quantization of linear sigma models

- Linear sigma models: M flat.
- $M = \mathbb{R}^n$ and $p \in \mathbb{R}^n$.
  $T_pM = T_p\mathbb{R}^n$: the tangent space of $M = \mathbb{R}^n$ at p.
- Heisenberg algebra: $\widehat{T_pM} = \widehat{T_pM}_- \oplus \widehat{T_pM}_0 \oplus \widehat{T_pM}_+$, where

$$\begin{aligned}
\widehat{T_pM}_- &= T_pM \otimes t^{-1}\mathbb{C}[t^{-1}], \\
\widehat{T_pM}_0 &= T_pM \otimes \mathbb{C}t^0 \oplus \mathbb{C}\mathbf{I} \simeq T_pM \oplus \mathbb{C}\mathbf{I}, \\
\widehat{T_pM}_+ &= TM \otimes t\mathbb{C}[t].
\end{aligned}$$

- Eigenfunctions $|\mathbf{k}\rangle = e^{ik_\mu x^\mu}$ of the Laplacian with eigenvalues $-k_\mu k^\mu$.

Free bosons: Quantization of linear sigma models

- Linear sigma models: M flat.
- $M = \mathbb{R}^n$ and $p \in \mathbb{R}^n$.
  $T_pM = T_p\mathbb{R}^n$: the tangent space of $M = \mathbb{R}^n$ at p.
- Heisenberg algebra: $\widehat{T_pM} = \widehat{T_pM}_- \oplus \widehat{T_pM}_0 \oplus \widehat{T_pM}_+$, where

$$\begin{aligned}
\widehat{T_pM}_- &= T_pM \otimes t^{-1}\mathbb{C}[t^{-1}], \\
\widehat{T_pM}_0 &= T_pM \otimes \mathbb{C}t^0 \oplus \mathbb{C}\mathbf{I} \simeq T_pM \oplus \mathbb{C}\mathbf{I}, \\
\widehat{T_pM}_+ &= TM \otimes t\mathbb{C}[t].
\end{aligned}$$

- Eigenfunctions $|\mathbf{k}\rangle = e^{ik_\mu x^\mu}$ of the Laplacian with eigenvalues $-k_\mu k^\mu$.

Vertex operator algebra and modules associated to free bosons

- Let $\widehat{T_pM}_+$ act on $|\mathbf{k}\rangle$ as 0, $\mathbf{I}$ on $|\mathbf{k}\rangle$ as 1 and elements of T_pM on $|\mathbf{k}\rangle$ as i times the vector fields obtained by parallel transporting the elements of T_pM. Then $\mathbb{C}|\mathbf{k}\rangle$ becomes a module for $\widehat{T_pM}_0 \oplus \widehat{T_pM}_+$.
- Induced module: $W_{|\mathbf{k}\rangle} = U(\widehat{T_pM}) \otimes_{U(\widehat{T_pM}_0 \oplus \widehat{T_pM}_+)} \mathbb{C}|\mathbf{k}\rangle$ (Fock space generated by $|\mathbf{k}\rangle$).
- $W_{|\mathbf{0}\rangle}$ is a vertex operator algebra and $W_{|\mathbf{k}\rangle}$ for any $\mathbf{k}$ is a module for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.
- The corresponding conformal field theory can be constructed by studying the correlation functions among these modules for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.

Vertex operator algebra and modules associated to free bosons

- Let $\widehat{T_pM}_+$ act on $|\mathbf{k}\rangle$ as 0, $\mathbf{I}$ on $|\mathbf{k}\rangle$ as 1 and elements of T_pM on $|\mathbf{k}\rangle$ as i times the vector fields obtained by parallel transporting the elements of T_pM. Then $\mathbb{C}|\mathbf{k}\rangle$ becomes a module for $\widehat{T_pM}_0 \oplus \widehat{T_pM}_+$.
- Induced module: $W_{|\mathbf{k}\rangle} = U(\widehat{T_pM}) \otimes_{U(\widehat{T_pM}_0 \oplus \widehat{T_pM}_+)} \mathbb{C}|\mathbf{k}\rangle$ (Fock space generated by $|\mathbf{k}\rangle$).
- $W_{|\mathbf{0}\rangle}$ is a vertex operator algebra and $W_{|\mathbf{k}\rangle}$ for any $\mathbf{k}$ is a module for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.
- The corresponding conformal field theory can be constructed by studying the correlation functions among these modules for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.

Vertex operator algebra and modules associated to free bosons

- Let $\widehat{T_pM}_+$ act on $|\mathbf{k}\rangle$ as 0, $\mathbf{I}$ on $|\mathbf{k}\rangle$ as 1 and elements of T_pM on $|\mathbf{k}\rangle$ as i times the vector fields obtained by parallel transporting the elements of T_pM. Then $\mathbb{C}|\mathbf{k}\rangle$ becomes a module for $\widehat{T_pM}_0 \oplus \widehat{T_pM}_+$.
- Induced module: $W_{|\mathbf{k}\rangle} = U(\widehat{T_pM}) \otimes_{U(\widehat{T_pM}_0 \oplus \widehat{T_pM}_+)} \mathbb{C}|\mathbf{k}\rangle$ (Fock space generated by $|\mathbf{k}\rangle$).
- $W_{|\mathbf{0}\rangle}$ is a vertex operator algebra and $W_{|\mathbf{k}\rangle}$ for any $\mathbf{k}$ is a module for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.
- The corresponding conformal field theory can be constructed by studying the correlation functions among these modules for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.

Vertex operator algebra and modules associated to free bosons

- Let $\widehat{T_pM}_+$ act on $|\mathbf{k}\rangle$ as 0, $\mathbf{I}$ on $|\mathbf{k}\rangle$ as 1 and elements of T_pM on $|\mathbf{k}\rangle$ as i times the vector fields obtained by parallel transporting the elements of T_pM. Then $\mathbb{C}|\mathbf{k}\rangle$ becomes a module for $\widehat{T_pM}_0 \oplus \widehat{T_pM}_+$.
- Induced module: $W_{|\mathbf{k}\rangle} = U(\widehat{T_pM}) \otimes_{U(\widehat{T_pM}_0 \oplus \widehat{T_pM}_+)} \mathbb{C}|\mathbf{k}\rangle$ (Fock space generated by $|\mathbf{k}\rangle$).
- $W_{|\mathbf{0}\rangle}$ is a vertex operator algebra and $W_{|\mathbf{k}\rangle}$ for any $\mathbf{k}$ is a module for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.
- The corresponding conformal field theory can be constructed by studying the correlation functions among these modules for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.

Sigma models on tori

- Given a torus, the vertex operator algebra is still $W_{|\mathbf{0}\rangle}$. But consider only modules $W_{|\mathbf{k}\rangle}$ for those $\mathbf{k}$ such that $e^{ik_\mu x^\mu}$ are well defined on the torus. Such $\mathbf{k}$ form a lattice.
- The corresponding conformal field theory can be constructed by studying the correlation functions among $W_{|\mathbf{k}\rangle}$ for such $\mathbf{k}$ for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.

Sigma models on tori

- Given a torus, the vertex operator algebra is still $W_{|\mathbf{0}\rangle}$. But consider only modules $W_{|\mathbf{k}\rangle}$ for those $\mathbf{k}$ such that $e^{ik_\mu x^\mu}$ are well defined on the torus. Such $\mathbf{k}$ form a lattice.
- The corresponding conformal field theory can be constructed by studying the correlation functions among $W_{|\mathbf{k}\rangle}$ for such $\mathbf{k}$ for the vertex operator algebra $W_{|\mathbf{0}\rangle}$.

First difficulty in generalizing this construction to the curved case

- M: a Riemannian manifold. $p \in M$. T_pM: the tangent space of $\mathbb{R}^n$ at p.
- Affinization $\widehat{T_pM}$.
- Problem: How to define an action of T_pM on smooth functions, in particular, on eigenfunctions of the Laplacian, at least near p.
- If we choose a local coordinate patch near p and define the action using derivatives with respect to the coordinates, the results depend on the local coordinate patch.
- If we define the action using the covariant derivatives, then we cannot view T_pM as an abelian Lie algebra becuase the commutators of covariant derivatives give the curvature tensor.

First difficulty in generalizing this construction to the curved case

- M: a Riemannian manifold. $p \in M$. T_pM: the tangent space of $\mathbb{R}^n$ at p.
- Affinization $\widehat{T_pM}$.
- Problem: How to define an action of T_pM on smooth functions, in particular, on eigenfunctions of the Laplacian, at least near p.
- If we choose a local coordinate patch near p and define the action using derivatives with respect to the coordinates, the results depend on the local coordinate patch.
- If we define the action using the covariant derivatives, then we cannot view T_pM as an abelian Lie algebra becuase the commutators of covariant derivatives give the curvature tensor.

First difficulty in generalizing this construction to the curved case

- M: a Riemannian manifold. $p \in M$. T_pM: the tangent space of $\mathbb{R}^n$ at p.
- Affinization $\widehat{T_pM}$.
- Problem: How to define an action of T_pM on smooth functions, in particular, on eigenfunctions of the Laplacian, at least near p.
- If we choose a local coordinate patch near p and define the action using derivatives with respect to the coordinates, the results depend on the local coordinate patch.
- If we define the action using the covariant derivatives, then we cannot view T_pM as an abelian Lie algebra becuase the commutators of covariant derivatives give the curvature tensor.

First difficulty in generalizing this construction to the curved case

- M: a Riemannian manifold. $p \in M$. T_pM: the tangent space of $\mathbb{R}^n$ at p.
- Affinization $\widehat{T_pM}$.
- Problem: How to define an action of T_pM on smooth functions, in particular, on eigenfunctions of the Laplacian, at least near p.
- If we choose a local coordinate patch near p and define the action using derivatives with respect to the coordinates, the results depend on the local coordinate patch.
- If we define the action using the covariant derivatives, then we cannot view T_pM as an abelian Lie algebra becuase the commutators of covariant derivatives give the curvature tensor.

First difficulty in generalizing this construction to the curved case

- M: a Riemannian manifold. $p \in M$. T_pM: the tangent space of $\mathbb{R}^n$ at p.
- Affinization $\widehat{T_pM}$.
- Problem: How to define an action of T_pM on smooth functions, in particular, on eigenfunctions of the Laplacian, at least near p.
- If we choose a local coordinate patch near p and define the action using derivatives with respect to the coordinates, the results depend on the local coordinate patch.
- If we define the action using the covariant derivatives, then we cannot view T_pM as an abelian Lie algebra becuase the commutators of covariant derivatives give the curvature tensor.

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $\widehat{TM}_-$: The vector bundle whose fiber at $p \in M$ is $\widehat{T_pM}_-$.
- $T(\widehat{TM}_-)$: The vector bundle whose fiber at $p \in M$ is $T(\widehat{T_pM}_-)$.
- $\Pi_U(T(\widehat{TM}_-))$: the space of parallel sections on U of the vector bundle $T(\widehat{TM}_-)$.

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $\widehat{TM}_-$: The vector bundle whose fiber at $p \in M$ is $\widehat{T_pM}_-$.
- $T(\widehat{TM}_-)$: The vector bundle whose fiber at $p \in M$ is $T(\widehat{T_pM}_-)$.
- $\Pi_U(T(\widehat{TM}_-))$: the space of parallel sections on U of the vector bundle $T(\widehat{TM}_-)$.

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $\widehat{TM}_-$: The vector bundle whose fiber at $p \in M$ is $\widehat{T_pM}_-$.
- $T(\widehat{TM}_-)$: The vector bundle whose fiber at $p \in M$ is $T(\widehat{T_pM}_-)$.
- $\Pi_U(T(\widehat{TM}_-))$: the space of parallel sections on U of the vector bundle $T(\widehat{TM}_-)$.

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $\widehat{TM}_-$: The vector bundle whose fiber at $p \in M$ is $\widehat{T_pM}_-$.
- $T(\widehat{TM}_-)$: The vector bundle whose fiber at $p \in M$ is $T(\widehat{T_pM}_-)$.
- $\Pi_U(T(\widehat{TM}_-))$: the space of parallel sections on U of the vector bundle $T(\widehat{TM}_-)$.

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $\widehat{TM}_-$: The vector bundle whose fiber at $p \in M$ is $\widehat{T_pM}_-$.
- $T(\widehat{TM}_-)$: The vector bundle whose fiber at $p \in M$ is $T(\widehat{T_pM}_-)$.
- $\Pi_U(T(\widehat{TM}_-))$: the space of parallel sections on U of the vector bundle $T(\widehat{TM}_-)$.

A sheaf of meromorphic open-string vertex algebras

Theorem (H. 2012)

The space $\Pi_U(T(\widehat{TM}_-))$ *has a natural structure of meromorphic open-string vertex algebra. The assignment* $U \to \Pi_U(T(\widehat{TM}_-))$ *gives a sheaf of meromorphic open-string vertex algebras.*

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Notations

- U: an open subset of M.
- $TM^{\mathbb{C}}$: The complexification of the tangent bundle of M.
- $T^m(TM^{\mathbb{C}})$: The vector bundle whose fibers are the m-th tensor powers of the complexifications of the tangent spaces of M.
- $T(TM^{\mathbb{C}})$: The vector bundle whose fibers are the tensor algebras of the complexifications of the tangent spaces of M.
- $\Gamma_U(T(TM^{\mathbb{C}}))$: the space of sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $\Pi_U(T(TM^{\mathbb{C}}))$: the space of parallel sections on U of the vector bundle $T(TM^{\mathbb{C}})$.
- $C^\infty(U)$: The space of smooth complex functions on U.
- $L(C^\infty(U))$: the algebra of linear operators on the space $C^\infty(U)$ of complex-valued smooth function on U.
- We define a linear map $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ as follows:

Definition of ϕ_U

- Given $f \in C^\infty(U)$, the m-th order covariant derivative $\nabla^m f$ can be viewed as a module map from the $C^\infty(U)$-module $\Gamma_U(T^m(TM^{\mathbb{C}}))$ to the $C^\infty(U)$-module $C^\infty(U)$. Since $\nabla^m f$ is linear in f, we can view ∇^m as a linear map from $C^\infty(U)$ to $\mathrm{Hom}_{C^\infty(U))}(\Gamma_U(T^m(TM^{\mathbb{C}})), C^\infty(U))$. Since such a map corresponds to a linear map from $\Gamma_U(T^m(TM^{\mathbb{C}}))$ to $L(C^\infty(U))$, we have a linear map

$$\psi_U^m : \Gamma_U(T^m(TM^{\mathbb{C}})) \to L(C^\infty(U))$$

corresponding to $\left(\sqrt{-1}\right)^m \nabla^m$. Thus we obtain

$$\psi_U : \Gamma_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U)).$$

- We define $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ to be the restriction of ψ_U^m to $\Pi_U(T(TM^{\mathbb{C}}))$.

Definition of ϕ_U

- Given $f \in C^\infty(U)$, the m-th order covariant derivative $\nabla^m f$ can be viewed as a module map from the $C^\infty(U)$-module $\Gamma_U(T^m(TM^{\mathbb{C}}))$ to the $C^\infty(U)$-module $C^\infty(U)$. Since $\nabla^m f$ is linear in f, we can view ∇^m as a linear map from $C^\infty(U)$ to $\mathrm{Hom}_{C^\infty(U))}(\Gamma_U(T^m(TM^{\mathbb{C}})), C^\infty(U))$. Since such a map corresponds to a linear map from $\Gamma_U(T^m(TM^{\mathbb{C}}))$ to $L(C^\infty(U))$, we have a linear map

$$\psi_U^m : \Gamma_U(T^m(TM^{\mathbb{C}})) \to L(C^\infty(U))$$

corresponding to $\left(\sqrt{-1}\right)^m \nabla^m$. Thus we obtain

$$\psi_U : \Gamma_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U)).$$

- We define $\phi_U : \Pi_U(T(TM^{\mathbb{C}})) \to L(C^\infty(U))$ to be the restriction of ψ_U^m to $\Pi_U(T(TM^{\mathbb{C}}))$.

A homomorphism of associative algebras

Example: Let g^{-1} be the element of $\Gamma_U(T^2(TM^{\mathbb{C}}))$ corresponding to the metric on the cotagent bundle. Then g^{-1} is parallel, that is, $g^{-1} \in \Pi_U(T^2(TM^{\mathbb{C}}))$ and $\phi_U(g^{-1}) = -\Delta$.

Theorem (H. 2012)

ϕ_U is a homomorphism of associative algebras from $\Pi_U(T(TM^{\mathbb{C}}))$ to $L(C^\infty(U))$.

A homomorphism of associative algebras

Example: Let g^{-1} be the element of $\Gamma_U(T^2(TM^{\mathbb{C}}))$ corresponding to the metric on the cotagent bundle. Then g^{-1} is parallel, that is, $g^{-1} \in \Pi_U(T^2(TM^{\mathbb{C}}))$ and $\phi_U(g^{-1}) = -\Delta$.

Theorem (H. 2012)

ϕ_U is a homomorphism of associative algebras from $\Pi_U(T(TM^{\mathbb{C}}))$ to $L(C^\infty(U))$.

Outline

Operator product expansion in quantum field theory

Nonlinear sigma models
- Classical nonlinear sigma models: harmonic maps
- Quantization (of linear sigma models)

A sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$
- Covariant derivatives, parallel tensor fields and smooth functions
- A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

The Lapacian as a component of a vertex operator

A presheaf

- By the theorem above, $C^\infty(U)$ is a $\Pi_U(T(TM^{\mathbb{C}}))$-module. For any $p \in M$, $\Pi_U(T(TM^{\mathbb{C}}))$ is isomorphic to the fixed point subspace $(T(T_pM^{\mathbb{C}}))^{H_p(U)}$ of $T(T_pM^{\mathbb{C}})$ under the holonormy group $H_p(U)$. Let

$$C_p(U) = T(T_pM^{\mathbb{C}}) \otimes_{(T(T_pM^{\mathbb{C}}))^{H_p(U)}} C^\infty(U).$$

- We know that $T(\widehat{T_pM_-})$ has a natural structure of meromorphic open-string vertex algebra and $T(\widehat{T_pM_-}) \otimes C_p(U)$ has a natural structure of left $T(\widehat{T_pM_-})$-module. Then $(T(\widehat{T_pM_-}))^{H_p(U)}$ is a meromorphic open-string vertex subalgebra of $T(\widehat{T_pM_-})$ and it is isomorphic to $\Pi_U(T(\widehat{TM_-}))$. In particular, $T(\widehat{T_pM_-}) \otimes C_p(U)$ is also a left $\Pi_U(T(\widehat{TM_-}))$-module.

A presheaf

- By the theorem above, $C^\infty(U)$ is a $\Pi_U(T(TM^{\mathbb{C}}))$-module. For any $p \in M$, $\Pi_U(T(TM^{\mathbb{C}}))$ is isomorphic to the fixed point subspace $(T(T_pM^{\mathbb{C}}))^{H_p(U)}$ of $T(T_pM^{\mathbb{C}})$ under the holonormy group $H_p(U)$. Let

$$C_p(U) = T(T_pM^{\mathbb{C}}) \otimes_{(T(T_pM^{\mathbb{C}}))^{H_p(U)}} C^\infty(U).$$

- We know that $T(\widehat{T_pM_-})$ has a natural structure of meromorphic open-string vertex algebra and $T(\widehat{T_pM_-}) \otimes C_p(U)$ has a natural structure of left $T(\widehat{T_pM_-})$-module. Then $(T(\widehat{T_pM_-}))^{H_p(U)}$ is a meromorphic open-string vertex subalgebra of $T(\widehat{T_pM_-})$ and it is isomorphic to $\Pi_U(T(\widehat{TM_-}))$. In particular, $T(\widehat{T_pM_-}) \otimes C_p(U)$ is also a left $\Pi_U(T(\widehat{TM_-}))$-module.

- Let W_U^0 be the left $\Pi_U(T(\widehat{TM}_-))$-submodule of $T(\widehat{T_pM}_-) \otimes C_p(U)$ generated by elements of the form $1 \otimes (1 \otimes_{(T(T_pM^{\mathbb{C}}))^{H_p(U)}} f)$ for $f \in C^\infty(U)$, where $1 \otimes_{(T(T_pM^{\mathbb{C}}))^{H_p(U)}} f$ is the image of $1 \otimes f$ under the projection from $T(T_pM^{\mathbb{C}}) \otimes C^\infty(U)$ to $C_p(U)$.
- The assignment $U \to W^0(U)$ gives a presheaf $\mathcal{W}^0$.

- Let W_U^0 be the left $\Pi_U(T(\widehat{TM}_-))$-submodule of $T(\widehat{T_pM}_-) \otimes C_p(U)$ generated by elements of the form $1 \otimes (1 \otimes_{(T(T_pM^{\mathbb{C}}))^{H_p(U)}} f)$ for $f \in C^\infty(U)$, where $1 \otimes_{(T(T_pM^{\mathbb{C}}))^{H_p(U)}} f$ is the image of $1 \otimes f$ under the projection from $T(T_pM^{\mathbb{C}}) \otimes C^\infty(U)$ to $C_p(U)$.
- The assignment $U \to W^0(U)$ gives a presheaf $\mathcal{W}^0$.

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

Let $\mathcal{W}$ be the sheafification of $\mathcal{W}^0$. The section of $\mathcal{W}$ on U is denoted W_U.

Theorem (H. 2012)

The space $W(U)$ of sections of $\mathcal{W}$ on U is a left $\Pi_U(T(\widehat{TM}_-))$-module and $\mathcal{W}$ is a sheaf of left modules for the sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras.

A sheaf $\mathcal{W}$ of left modules for the sheaf $\mathcal{V}$

Let $\mathcal{W}$ be the sheafification of $\mathcal{W}^0$. The section of $\mathcal{W}$ on U is denoted W_U.

Theorem (H. 2012)

The space $W(U)$ of sections of $\mathcal{W}$ on U is a left $\Pi_U(T(\widehat{TM}_-))$-module and $\mathcal{W}$ is a sheaf of left modules for the sheaf $\mathcal{V}$ of meromorphic open-string vertex algebras.

The Lapacian as a component of a vertex operator

Many conjectures in geometry can be obtained from quantum field theory by interpreting some geometric or analytic objects as quantum-field-theoretic objects. Here we give an example of such an interpretation. The applications are still to be carried out.

Let $\{E_i\}_{i=1}^n$ be an orthonormal frame in an open subset U of M. Let

$$g_{\mathbb{C}}^{-1}(-1,-1) = \sum_{i=1}^{n} (E_i \otimes_{\mathbb{R}} t^{-1}) \otimes (E_i \otimes_{\mathbb{R}} t^{-1}) \in \Pi_U(T^2(\widehat{TM}_-)).$$

Then $g_{\mathbb{C}}^{-1}(-1,-1)$ is in fact well defined on any open subset of M. In particular, it is well defined on M. Also, we can identify $C^\infty(M)$ with a subspace of W_M.

Theorem (H. 2012)

The coefficient of the x^{-2} term of $Y_{W_M}(-g_{\mathbb{C}}^{-1}(-1,-1), x)$ acting on $f \in C^\infty(M)$ is equal to Δf.

The Lapacian as a component of a vertex operator

Many conjectures in geometry can be obtained from quantum field theory by interpreting some geometric or analytic objects as quantum-field-theoretic objects. Here we give an example of such an interpretation. The applications are still to be carried out.

Let $\{E_i\}_{i=1}^n$ be an orthonormal frame in an open subset U of M. Let

$$g_{\mathbb{C}}^{-1}(-1,-1) = \sum_{i=1}^{n} (E_i \otimes_{\mathbb{R}} t^{-1}) \otimes (E_i \otimes_{\mathbb{R}} t^{-1}) \in \Pi_U(T^2(\widehat{TM}_-)).$$

Then $g_{\mathbb{C}}^{-1}(-1,-1)$ is in fact well defined on any open subset of M. In particular, it is well defined on M. Also, we can identify $C^\infty(M)$ with a subspace of W_M.

Theorem (H. 2012)

The coefficient of the x^{-2} term of $Y_{W_M}(-g_{\mathbb{C}}^{-1}(-1,-1), x)$ acting on $f \in C^\infty(M)$ is equal to Δf.

The Lapacian as a component of a vertex operator

Many conjectures in geometry can be obtained from quantum field theory by interpreting some geometric or analytic objects as quantum-field-theoretic objects. Here we give an example of such an interpretation. The applications are still to be carried out.

Let $\{E_i\}_{i=1}^n$ be an orthonormal frame in an open subset U of M. Let

$$g_{\mathbb{C}}^{-1}(-1,-1) = \sum_{i=1}^{n} (E_i \otimes_{\mathbb{R}} t^{-1}) \otimes (E_i \otimes_{\mathbb{R}} t^{-1}) \in \Pi_U(T^2(\widehat{TM}_-)).$$

Then $g_{\mathbb{C}}^{-1}(-1,-1)$ is in fact well defined on any open subset of M. In particular, it is well defined on M. Also, we can identify $C^\infty(M)$ with a subspace of W_M.

Theorem (H. 2012)

The coefficient of the x^{-2} term of $Y_{W_M}(-g_{\mathbb{C}}^{-1}(-1,-1), x)$ acting on $f \in C^\infty(M)$ is equal to Δf.

- The construction generalizes without difficulties to differential forms.
- We expect that when M is Ricci flat or Kähler or Calabi-Yau, we have stronger results.
- The main challenge is to construct correlation functions from all modules for the meromorphic open-string vertex algebra associated to M and prove that they satisfy all the properties for correlation functions for a quantum field theory, in particular, the operator product expansion.

- The construction generalizes without difficulties to differential forms.
- We expect that when M is Ricci flat or Kähler or Calabi-Yau, we have stronger results.
- The main challenge is to construct correlation functions from all modules for the meromorphic open-string vertex algebra associated to M and prove that they satisfy all the properties for correlation functions for a quantum field theory, in particular, the operator product expansion.

- The construction generalizes without difficulties to differential forms.
- We expect that when M is Ricci flat or Kähler or Calabi-Yau, we have stronger results.
- The main challenge is to construct correlation functions from all modules for the meromorphic open-string vertex algebra associated to M and prove that they satisfy all the properties for correlation functions for a quantum field theory, in particular, the operator product expansion.